Federated LoRA Adaptation of BiomedCLIP

Across Four International Chest X-Ray Cohorts

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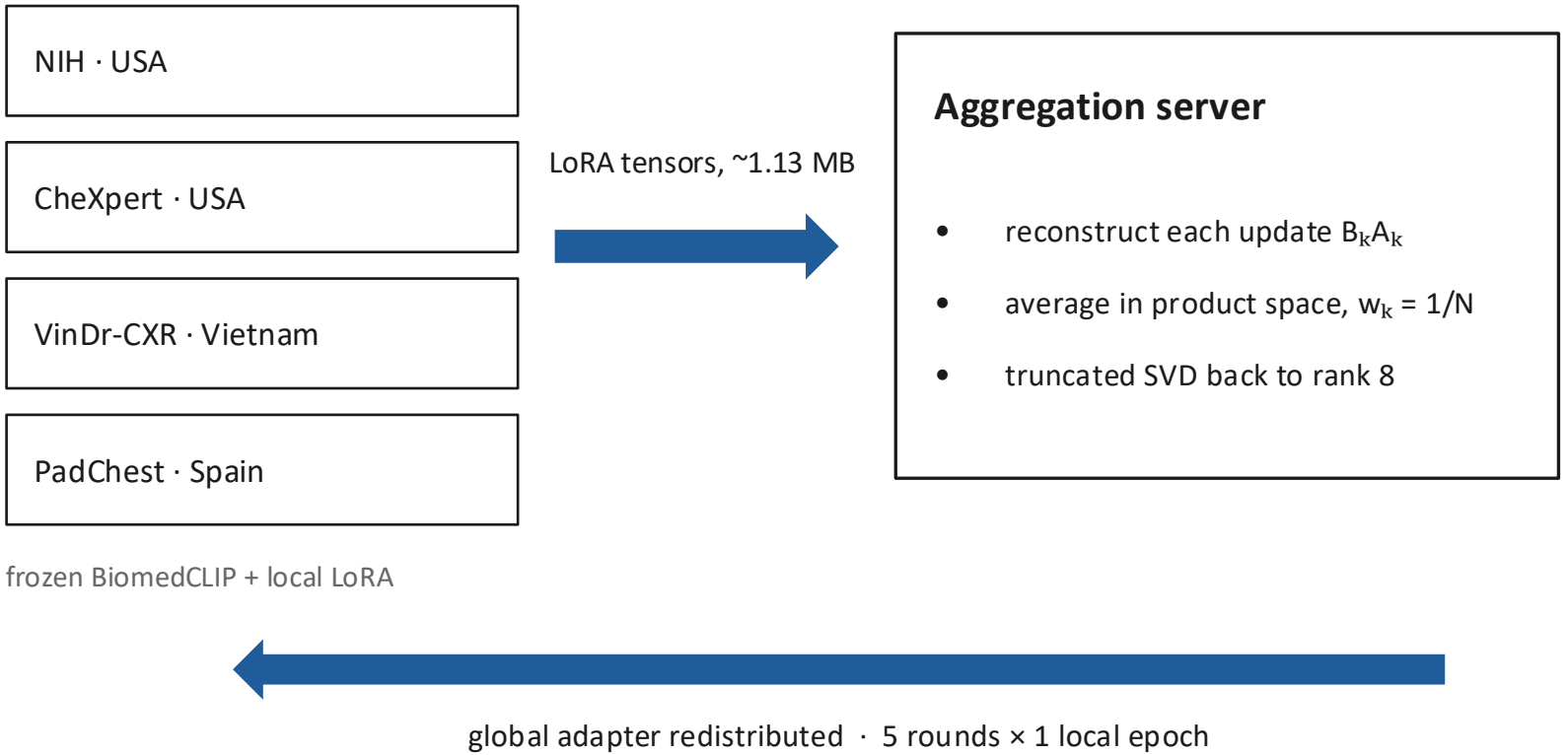
github.com/GaireLaboratory/FedLoRA-BiomedCLIP

Motivation

- Privacy regulation keeps chest radiographs inside the institution that acquired them, so any single site trains on a narrow slice of the population.
- Federated learning is the standard answer: train a shared model, keep raw images local, exchange only model updates.
- Biomedical vision-language models such as BiomedCLIP are a natural fit — prompt-based labels handle cohorts with different label vocabularies, where a fixed classification head cannot.
- But federated-LoRA methods have been validated almost entirely on natural-language tasks, and prior federated chest X-ray work uses CNNs with fixed label heads.

What happens when the local objective is image–text contrastive alignment and clients differ by scanner, protocol, geography and label vocabulary? That is what we benchmark.

Setup



- BiomedCLIP: ViT-B/16 image encoder, frozen PubMedBERT text encoder
- LoRA on the fused qkv projection, r = 8, α = 16, dropout 0.1
- ~0.25% of parameters trainable; >99% less traffic than full-model exchange
- Raw images never leave their source cohort

Averaging factors is not averaging updates

Each client's LoRA update is the product B_kA_k of two small matrices.

Naive — what plain FedAvg does

average the A's and the B's separately, then multiply: mean(B) × mean(A) ≠ mean(BA)

0.684

the merged update no longer represents what any client learned

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Product space — FlexLoRA, and what we use

reconstruct each B_kA_k, average those, then project back to rank 8 with a truncated SVD

0.781

four rank-8 updates can sum to rank 32, so this is an approximation — it retains 94.6% of the averaged update's singular-value energy (92.9–95.4% across the twelve blocks)

+0.097 AUC Naive merging lands at 0.684 — the level of the untrained backbone (0.687).

Four cohorts, three continents

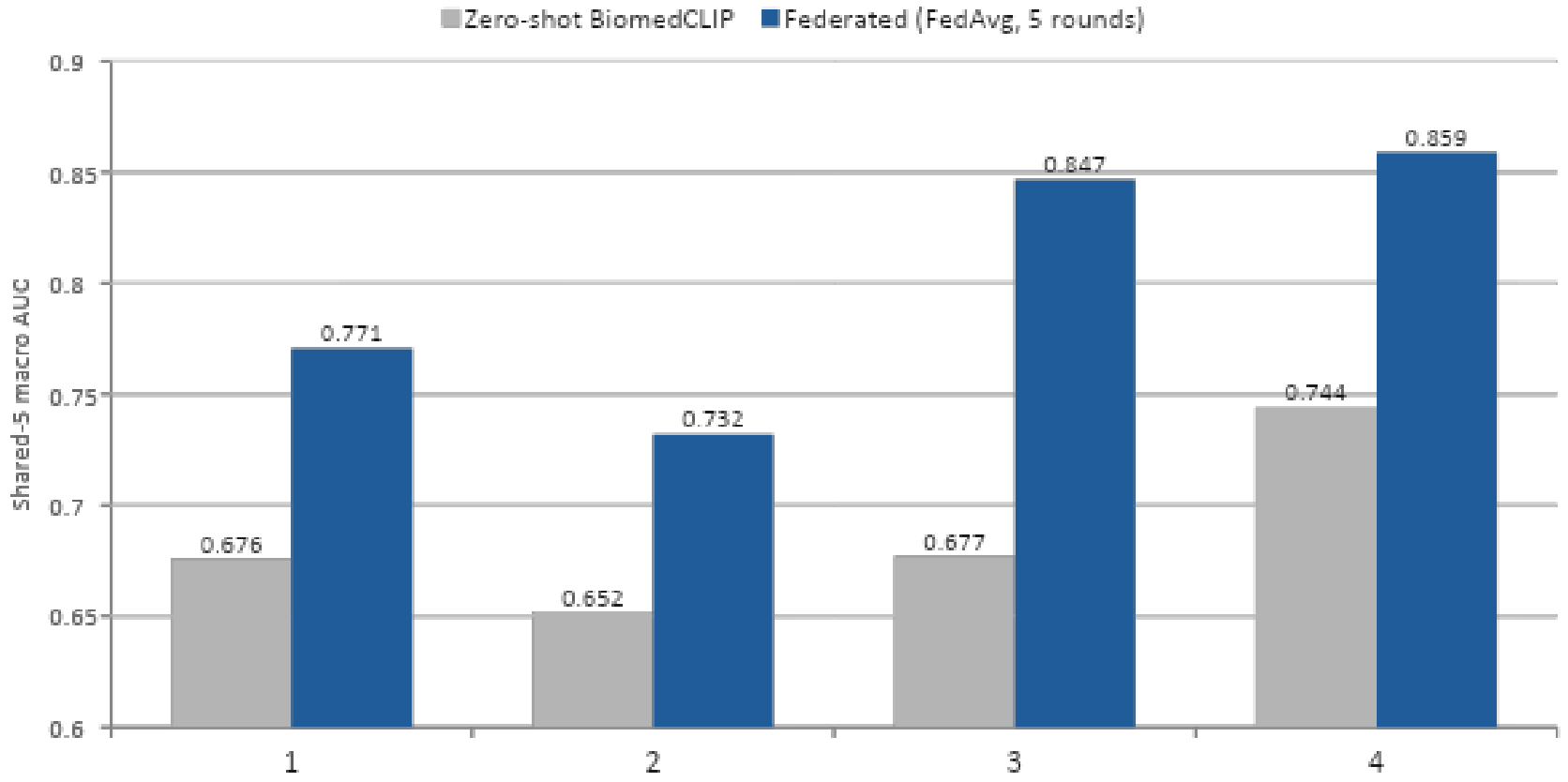
Real institutional heterogeneity, not a simulated partition of one dataset.

Cohort	Region	Train	Val	Test	Split
NIH ChestX-ray14	USA	77,872	8,652	25,596	Patient-level (official)
CheXpert	USA	153,182	18,958	18,729	Patient-level
VinDr-CXR	Vietnam	12,000	1,500	1,500	Image-level
PadChest	Spain	65,173	8,078	8,147	Patient-level

Three experiments of increasing collaboration

1. Single-client — each cohort trains alone — **0.776** mean shared-5 macro AUC
2. One-shot merge — the four adapters merged once — **0.781**
3. Multi-round federation — five rounds of train–aggregate — **0.802**

Federation improves every cohort



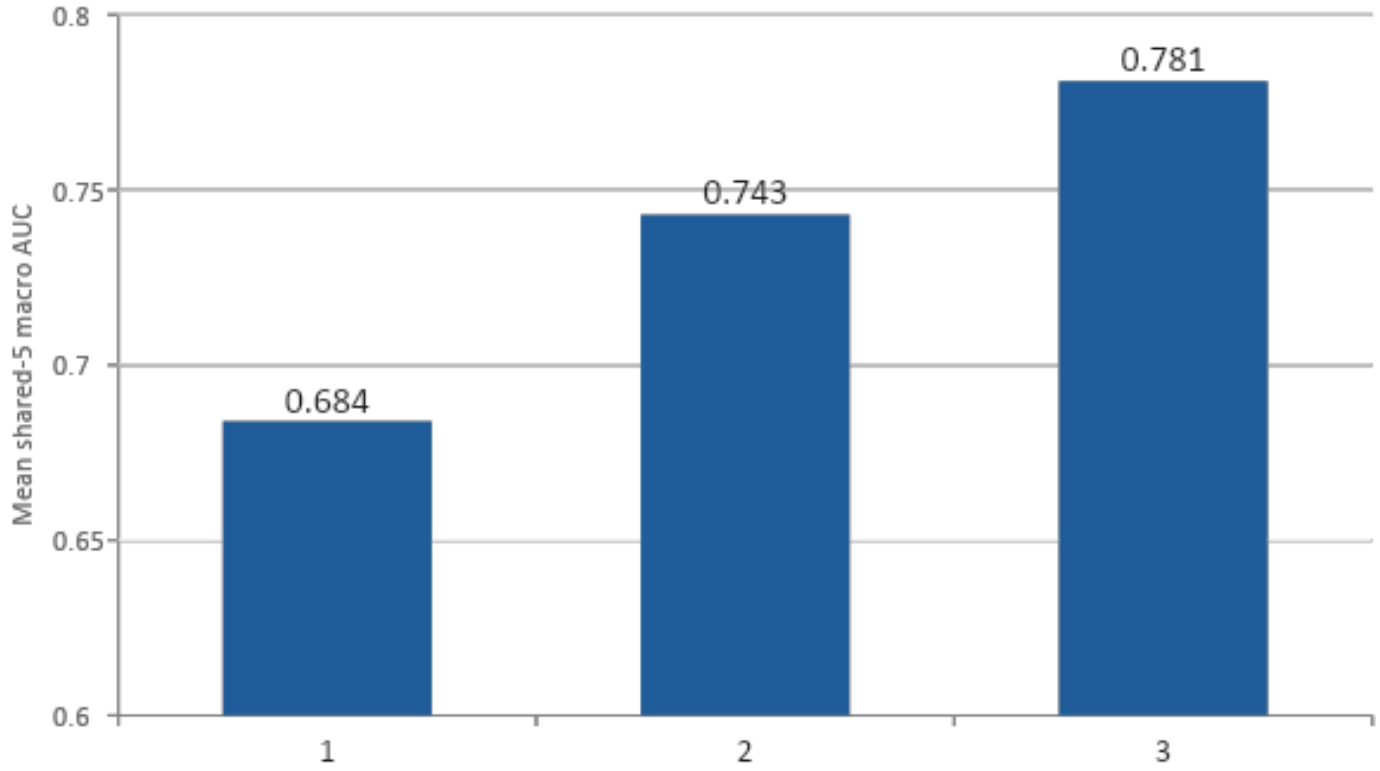
Gain over the unadapted backbone

NIH	**+0.095**
CheXpert	**+0.080**
VinDr	**+0.170**
PadChest	**+0.115**
Mean	**+0.115**

Federation also exceeds the best single-client model (0.776) and approaches a centralized reference (0.812) trained on pooled data under a different budget.

The merge rule decides everything

One-shot aggregation of the four single-client adapters — the four inputs are identical, only the merge differs.



Equal client weights

We aggregate with $w_k = 1/N$ rather than the data-size weighting of the original FedAvg. Size weighting gives CheXpert — 153k images and our weakest single-client model — roughly half the merge weight, costing 0.038 AUC. We state this deviation explicitly in the paper.

Scope

This comparison is one-shot only. Verifying that the naive–SVD gap persists across multiple rounds is left to future work.

Why is our largest cohort our weakest model?

In contrastive training every negative image gets the identical caption, so the CLIP loss pushes apart pairs that should match. We counted these collisions per batch.

Cohort	Caption-collision rate	Single-client mean AUC
CheXpert	**0.33**	**0.618**
PadChest	0.61	0.774
NIH	0.77	0.776
VinDr-CXR	0.78	0.740

The hypothesis fails — and that is the finding

CheXpert has the lowest collision rate of the four cohorts yet the worst model, while NIH and VinDr have the highest rates and train well. If collisions were dominant, this ordering would be reversed.

What we conclude

CheXpert's difficulty more plausibly reflects its uncertain-label policy and documented label-quality issues.

We rule collisions out as the explanation for CheXpert specifically. We do not claim they are harmless in general, and we do not isolate the true cause.

Takeaways

- Biomedical vision-language models federate well. BiomedCLIP with LoRA improves on all four cohorts over the unadapted backbone (0.687 → 0.802) and approaches a centralized reference (0.812) without pooling any data.
- The merge rule is not an implementation detail. Naive factor averaging costs 0.097 AUC and lands at the level of the untrained backbone; product-space SVD aggregation with equal client weights recovers it.
- FedProx adds nothing over FedAvg (0.799 vs 0.802). LoRA constrains the update subspace, and the frozen text encoder means every client aligns to the same fixed prompt embeddings.
- The benchmark itself, rather than a new algorithm, is our contribution — four real cohorts on three continents, with code released.

Limitations: single seed, fixed rank r = 8, and the naive-vs-SVD ablation is one-shot only — small gaps should be read as indicative.

Thank you!

Questions & Discussion